

Gait Biomechanics in Anterior Cruciate Ligament–reconstructed Knees at Different Time Frames Postsurgery

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ABSTRACT

GOETSCHUIS, J., J. HERTEL, S. A. SALIBA, S. F. BROCKMEIER, and J. M. HART. Gait Biomechanics in Anterior Cruciate Ligament–reconstructed Knees at Different Time Frames Postsurgery. *Med. Sci. Sports Exerc.*, Vol. 50, No. 11, pp. 2209–2216, 2018. **Purpose:** To examine knee and hip biomechanics during walking and jogging in groups of ACLR patients at early, mid, and late time frames postsurgery and healthy controls. **Methods:** Participants included individuals with a history of primary, unilateral ACLR, stratified into early (1.4 ± 0.4 yr post, $n = 18$), mid (3.3 ± 0.6 yr post, $n = 20$), and late (8.5 ± 2.8 yr post, $n = 20$) ACLR groups based on time postsurgery, and a healthy control group ($n = 20$). Walking and jogging motion capture analysis of knee and hip kinetics and kinematics were measured in the sagittal and frontal planes. Interlimb (within groups) and between-group comparisons were performed for all gait variables. Statistical comparisons were made across the gait cycle by plotting graphs of means and 90% confidence intervals and identifying regions of the gait cycle in which the 90% confidence intervals did not overlap. **Results:** Early ACLR group demonstrated reduced knee flexion, knee extension, knee adduction, and hip adduction moments on the ACLR limb. Mid ACLR group demonstrated no gait differences between limbs or other groups. Late ACLR group demonstrated reduced knee flexion moments, and greater knee and hip adduction moments in their ACLR limb. Control group demonstrated no interlimb differences. **Conclusions:** Walking and jogging gait biomechanics presented differently in patients at different stages in time after ACLR surgery. The early ACLR group demonstrated lower sagittal and frontal plane joint loading on the ACLR limb compared with contralateral and control limbs. The mid ACLR group did not demonstrate any gait differences compared with the contralateral or control limb. The late ACLR group demonstrated lower sagittal plane joint loading compared with control limbs and greater frontal plane joint loading compared to contralateral and control limbs. **Key Words:** WALK, JOG, KNEE, HIP, FLEXION MOMENT, ADDUCTION MOMENT

Anterior cruciate ligament (ACL) reconstruction (ACLR) surgery in combination with postsurgical rehabilitation is a common treatment for physically active patients wishing to return to sport and exercise after ACL injury (1). An estimated 130,000 ACLR surgeries are performed each year in the United States and evidence suggest that ACLR surgery rates are increasing annually (2). Current clinical care after ACLR is focused on the goal of a safe and timely return to sport and exercise (3); however, evidence suggests that patients with a history of ACLR commonly report long-term knee dysfunction (4–6), lower

physical activity levels (6), and are at a high risk for early onset posttraumatic osteoarthritis (PTOA) (5,7). A recent systematic review reported that up to 48% of ACLR patients had evidence of knee OA within their second decade postsurgery (7). This is particularly concerning when considering that the highest rates of ACLR are performed in patients under the age of 20 yr (8).

Persistent adaptations in lower extremity joint mechanics during repetitive activities such as walking and jogging may contribute to the development of PTOA after ACLR (9). Unresolved impairments in proprioception (10), muscle function (11), and movement coordination (12) may manifest into abnormal lower extremity joint motion and loading during bipedal activities of daily living and exercise (13). Even small deviations in normal knee joint loading between limbs may lead to altered wear patterns on the articular tissues of the joint, and over time and repetition, these forces could cause rapid or higher magnitude joint degeneration (14). Despite an emphasis on restoring lower-extremity strength and movement symmetry during postsurgical rehabilitation, evidence suggests that patients may continue to demonstrate changes in joint loading patterns for years postsurgery (15).

Adaptations in sagittal and frontal plane kinetics have been indicated as potential signs of unhealthy mechanical

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loading on the knee joint during gait (15). Reductions in external knee flexion moments during the stance phase of gait have been reported in the ACL injured and ACLR knees (15,16). This “quadriceps avoidance” gait pattern is thought to occur in response to quadriceps muscle dysfunction (17), and evidence has shown that ACLR patients with weak quadriceps demonstrate a reduced external knee flexion moment compared to uninjured controls (18). Additionally, research has focused on the external knee adduction moment in patients with a history of ACLR (19–22) based on evidence suggesting that the knee adduction loading pattern may be related to tibiofemoral contact forces and the progression of osteoarthritis (23,24).

Posttraumatic osteoarthritis is a condition that develops over time after ACLR (7). Despite numerous studies examining joint loading patterns during gait in ACLR knees, there is still limited understanding of how gait patterns present over the long-term course of time postsurgery. Conventional theory suggests that although patients do experience gait improvements over time, some level of abnormal joint loading may persist long after ACLR. Two longitudinal studies have supported this theory, both observing improvements in knee sagittal (25) and frontal (21) plane loading that never reached normal symmetry after 2 to 3 yr postsurgery. Although these two longitudinal studies are some of the best available evidence regarding changes in joint loading over time, the 2- to 3-yr follow-up times in those studies may be too early to extrapolate to long-term gait adaptations. Abnormal knee joint loading in the sagittal (26,27) and frontal (19,21,28) planes have been observed in samples of ACLR patients averaging 3 to 5 yr postsurgery; however, the majority of ACLR samples included participants across a wide spectrum of times postsurgery, combining patient just returning to activity with those 10+ yr postsurgery, thus limiting our true understanding of joint loading patients at later time frames postsurgery. To better understand the theoretical relationship between posttraumatic adaptations in gait and the development of PTOA we need to be more intentional about understanding how lower extremity gait patterns present over the course of time postsurgery. Therefore, the purpose of this study was to evaluate and compare the presence of abnormal knee and hip joint biomechanics during walking and jogging in groups of individuals at

early, mid, and late time frames after unilateral ACLR surgery and a group of healthy controls.

METHODS

Data collection was performed in a laboratory setting and included three-dimensional motion capture of participants' gait during treadmill walking and jogging. Primary gait variables included knee and hip kinetics and kinematics in the sagittal and frontal planes and vertical ground reaction forces (vGRF). Comparisons were made between limbs, involved versus uninvolved, within each group and between the involved limb of all four groups, early ACLR, mid ACLR, late ACLR, and healthy controls. For ACLR participants, the surgical limb was treated as the involved limb. For control participants, one limb was randomly selected using a coin flip as the “involved limb” for all interlimb comparisons in an effort to remove limb selection bias. This study was approved by our university's institutional review board for health sciences research and all participants provided written informed consent.

Participants. A total of seventy-six individuals between the ages of 18 and 35 yr were recruited from our local university community for participation in this study. Fifty-six individuals had a history of primary, unilateral ACLR, were greater than 9 months postsurgery and had returned to exercise/sport with no physical activity restrictions imposed by their healthcare providers (Table 1). The ACLR participants were excluded if they had a history of multiligament knee surgery, surgical complications, or bilateral knee joint surgery. There were no restrictions on participation based on ACLR graft type or a history of meniscectomy or repair at the time of ACLR; however, participants with active meniscal symptoms (joint line pain or clicking/catching) were excluded. The ACLR participants were stratified into groups based on their time postsurgery: early ACLR: 6 to 24 months ($n = 18$), mid ACLR: 2 to 5 yr ($n = 20$), late ACLR: 5 to 15 yr ($n = 18$). Twenty recreationally active individuals with no history of lower extremity injury or surgery and no current symptoms of lower extremity pain or neuropathy participated as healthy controls.

Data collection. Demographic variables, including age, sex, mass, height, and time postsurgery, were collected, and all

TABLE 1. Early, mid, and late ACLR and control groups demographics.

	Early ACLR ($n = 18$)	Mid ACLR ($n = 20$)	Late ACLR ($n = 18$)	Control ($n = 20$)
Sex	11 F, 7 M	16 F, 4 M	12 F, 6 M	13 F, 7 M
Age (yr)	21.7 $\pm$ 4.1	20.5 $\pm$ 2.2	26.7 $\pm$ 4.4*	22.4 $\pm$ 3.2
Mass (kg)	68.7 $\pm$ 15.6	68.5 $\pm$ 9.9	69.5 $\pm$ 12.7	68.9 $\pm$ 13.1
Height (m)	1.72 $\pm$ 0.12	1.73 $\pm$ 0.09	1.73 $\pm$ 0.10	1.71 $\pm$ 0.13
IKDC (0–100)	85.4 $\pm$ 9.2**	86.0 $\pm$ 7.2**	89.8 $\pm$ 10.1**	99.5 $\pm$ 2.0
Godin	69.2 $\pm$ 22.8	74.5 $\pm$ 13.0	59.9 $\pm$ 26.4	70.5 $\pm$ 17.9
Time postsurgery (yr)	1.4 $\pm$ 0.4***	3.3 $\pm$ 0.6***	8.6 $\pm$ 2.8***	NA
Graft type	Patella tendon—7 Hamstrings—10 Cadaver—1	Patella tendon—11 Hamstrings—5 Cadaver—4	Patella tendon—10 Hamstrings—5 Cadaver—3	NA

*Significantly greater than all other groups ($P < 0.001$).

**Significantly lower than control group ($P < 0.001$).

***Significantly different from other ACLR groups ($P < 0.001$).

F, female; M, male; IKDC, International Knee Documentation Committee Subjective Knee Evaluation Score; Godin, Godin leisure-time activity score.

participants completed the International Knee Documentation Committee subjective knee evaluation (29) and the Godin leisure time exercise questionnaire (30) as descriptors of self-reported knee function and physical activity levels.

The motion capture system included Motion Monitor (Innovative Sports Training, Inc., Chicago, IL) and Nexus (Vicon Motion Systems, Ltd, UK) softwares, 12 Bonita-10 cameras (Vicon Motion Systems, Ltd, UK) and a split-belt instrumented treadmill (Bertec Corporation, Columbus, OH). Retro-reflective markers (14 mm) were placed bilaterally on the heel, second toe, lateral malleoli, lateral shank, lateral knee joint, lateral thigh, anterolateral thigh, and anterior superior iliac spine (ASIS). A four-marker sacral cluster was secured around the pelvis and aligned with the sacrum. Static calibration trials were performed standing with feet-shoulder width apart and toes facing forward. Each body segment was defined using a minimum of three markers: foot: heel, second toe, lateral malleoli; shank: lateral malleoli, lateral shank, lateral knee joint; thigh: lateral knee joint, lateral thigh, anterolateral thigh; and pelvis: left and right ASIS, sacral cluster. Ankle and knee joint centers were defined using anthropometric measures. The hip joint was defined using ASIS markers and the Bell method (31). Participants performed trials in their preferred shoes used for jogging exercise. Walking and jogging trials were performed at standardized speeds of $1.34 \text{ m}\cdot\text{s}^{-1}$ (3.0 mph) and $2.68 \text{ m}\cdot\text{s}^{-1}$ (6.0 mph), respectively, with 5:00 min warm-up periods for each task prior to collection. Participants ambulated continuously while ten sequential capture periods of 3 s each were collected during walking and jogging.

Data processing. Kinematic data were sampled at 100 Hz, treadmill force plate data were sampled at 1000 Hz, and all data were smoothed using a low-pass 20-Hz Butterworth filter. Kinetic and kinematics variables were reduced to 100 data points representing 1% to 100% of the gait cycle (heel contact to heel contact). Variables were calculated using the average of 10 strides for each limb. For each capture period, the first full gait cycle on the involved and uninvolved limb were selected using a threshold of 20 N to define heel contacts. Vertical ground reaction forces were reported in newtons normalized by body mass ($\text{N}\cdot\text{kg}^{-1}$), kinematic variables were reported in rotational degrees, and kinetic variables were reported in external moments, newton-meters ($\text{N}\cdot\text{m}$) normalized to body mass and height ($\text{N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$). All gait data was processed using the Motion Monitor software.

Statistical analysis. Demographics and questionnaire data were compared between each group using one-way ANOVA, Tukey's LSD, and χ^2 analyses (sex, graft-type). Each gait variable was compared between the involved and uninvolved limbs within each group, between the involved limbs of each ACLR group and both healthy limbs of the control group, and between the involved limb of each ACLR group. Comparisons were made across the entire gait cycle by plotting means and 90% confidence interval (CI) ensemble curves for each 1% of the gait cycles. Kinematic variables were analyzed across 1% to 100% of the gait cycle. Kinetics variables were only analyzed across 1% to 60% and 1% to

40% of the gait cycle for walking and jogging, respectively. Statistically significant differences between limbs or groups were identified by examining the plots and data for regions of the gait cycle where the 90% CI curves did not overlap for a minimum of three consecutive percentages (3%) of the gait cycle (32,33). Within these regions of the gait cycle, the average mean and standard deviation across the region for each limb/group and average Cohen's *d* effect sizes (34) and 95% CI between limbs/groups were calculated. Results were interpreted using the following gait cycle phases for walking (1%–10%, loading-response; 11%–30%, mid-stance; 31%–50%, terminal-stance; 51%–60%, preswing; 61%–73%, initial-swing; 74%–87%, mid-swing; 88%–100%, terminal swing) and jogging (1%–40%, stance; 41%–100%, swing). Data from two participants were not included in the jogging analyses: one participant from the mid ACLR group ($n = 19$) did not want to attempt the jogging task and one participant from the late ACLR group ($n = 17$) lost a marker during collection, and data were unable to be processed.

RESULTS

Group demographics and questionnaire results are presented in Table 1. Summary tables containing the results of the interlimb comparisons and ACLR versus control limb comparisons are presented in Tables 2 and 3, respectively. Mean ensemble curves for the knee sagittal, knee frontal, and hip frontal moments are presented for all three ACLR groups during walking (Fig. 1) and jogging (Fig. 2). Additional figures, including the mean and 90% CI ensemble curves for all interlimb and between group comparisons, are presented in the Appendix (Figures 1–18, Supplemental Digital Content 1, <http://links.lww.com/MSS/B313>).

Interlimb comparisons within groups. While walking, the early ACLR group demonstrated lower vGRF during preswing, lower external knee extension moments during terminal stance, greater knee flexion motion during pre-swing, lower external knee adduction moments during preswing, and greater hip abduction motion during mid and terminal swing on the involved limb compared with the uninvolved limb. While jogging, the early ACLR group demonstrated lower vGRF during stance, lower external knee flexion moments during stance, less knee flexion during stance, lower external knee adduction moments during stance, lower external hip adduction moments during stance, and less hip adduction motion during swing on the involved limb compared with the uninvolved limb.

There were no significant interlimb differences in walking or jogging kinetic or kinematic variables in the mid ACLR group.

While walking, the late ACLR group demonstrated greater external knee adduction moments during mid-stance and greater external hip adduction moments during mid-stance through pre-swing on the involved limb compared to the uninvolved limb. During the stance phase of jogging, the late

TABLE 2. Significant kinetic and kinematic interlimb differences within ACLR and control groups.

Kinetic and Kinematic Variables	Gait Cycle Region	Involved Mean $\pm$ SD	Uninvolved Mean $\pm$ SD	Effect Size and 95% CI
Walking differences				
Early ACLR				
vGRF	52%–56%	6.3 $\pm$ 1.8	7.1 $\pm$ 1.7	0.95 [0.26–1.63]
External knee extension moment	46%–51%	0.15 $\pm$ 0.11	0.25 $\pm$ 0.11	0.85 [0.16–1.53]
Knee flexion motion	47%–59%	14.8 $\pm$ 5.1	11.4 $\pm$ 8.9	0.86 [0.17–1.54]
External knee adduction moment	52%–58%	0.14 $\pm$ 0.09	0.22 $\pm$ 0.08	0.98 [0.28–1.67]
Hip abduction motion	82%–95%	5.2 $\pm$ 2.7	3.1 $\pm$ 2.2	0.86 [0.18–1.54]
Mid ACLR (No differences)				
Late ACLR				
External knee adduction moment	16%–32%,	0.33 $\pm$ 0.11	0.24 $\pm$ 0.09	0.87 [0.19–1.56]
External hip adduction moment	10%–58%	0.48 $\pm$ 0.15	0.30 $\pm$ 0.17	1.13 [0.42–1.84]
Control (No differences)				
Jogging differences				
Early ACLR				
vGRF	19%–24%	19.2 $\pm$ 1.4	20.6 $\pm$ 1.2	1.10 [0.40–1.80]
External knee flexion moment	10%–20%	0.67 $\pm$ 0.34	0.93 $\pm$ 0.22	0.91 [0.22–1.59]
Knee flexion motion	19%–21%	28.3 $\pm$ 4.2	32.3 $\pm$ 5.1	0.85 [0.17–1.54]
External knee adduction moment	23%–25%,	0.33 $\pm$ 0.23	0.53 $\pm$ 0.20	0.92 [0.23–1.61]
External hip adduction moment	23%–25%,	0.59 $\pm$ 0.38	0.88 $\pm$ 0.29	0.85 [0.17–1.53]
Hip adduction motion	86%–96%	1.2 $\pm$ 3.4	3.8 $\pm$ 2.5	0.87 [0.18–1.55]
Mid ACLR (No differences)				
Late ACLR				
External hip adduction moment	7%–9%, 13%–16%, 20%–23%, 27%–30%	1.00 $\pm$ 0.41	0.61 $\pm$ 0.36	1.01 [0.29–1.72]
Control (No differences)				

vGRF, N·m·kg⁻¹; external moments, N·m·kg⁻¹·m⁻¹; motions, °.

Gait cycle region, region of gait cycle that significant difference was observed.

ACLR group demonstrated greater external hip adduction moments on the involved limb compared with the uninvolved.

There were no significant interlimb differences in walking or jogging kinetic or kinematic variables in the control group.

Comparisons between ACLR groups and control group. While walking, the early ACLR group demonstrated lower vGRF during preswing, lower external knee extension moments during terminal stance and greater knee flexion during terminal stance through preswing compared to the healthy controls. During the stance phase of jogging, the early

ACLR group demonstrated lower vGRF, lower external knee flexion moments, less knee flexion motion, and a lower external knee adduction moment compared with the healthy controls.

There were no significant differences between the mid ACLR group and the healthy controls during walking or jogging.

While walking, the late ACLR group demonstrated greater vGRF during terminal stance, lower external knee flexion moments during mid stance, less knee abduction motion from loading-response through terminal stance and

TABLE 3. Significant kinetic and kinematic differences between ACLR groups and the control group.

Kinetic and Kinematic Variables	Gait Cycle Region	ACLR Mean $\pm$ SD	Control Mean $\pm$ SD	Effect Size and 95% CI
Walking differences				
Early ACLR				
vGRF	53%–57%	6.3 $\pm$ 1.2	7.1 $\pm$ 0.9	0.77 [0.20–1.35]
External knee extension moment	44%–53%	0.15 $\pm$ 0.11	0.23 $\pm$ 0.08	1.01 [0.42–1.59]
Knee flexion motion	44%–60%	14.7 $\pm$ 5.2	11.4 $\pm$ 3.4	0.84 [0.27–1.42]
Mid ACLR (No differences)				
Late ACLR				
vGRF	34%–37%	7.4 $\pm$ 0.4	7.1 $\pm$ 0.4	0.78 [0.20–1.35]
External knee flexion moment	14%–17%	0.24 $\pm$ 0.10	0.31 $\pm$ 0.10	0.90 [0.30–1.49]
Knee abduction motion	1%–52%, 96%–100%	0.1 $\pm$ 2.7	2.2 $\pm$ 2.5	0.85 [0.27–1.43]
Hip abduction motion	74%–89%	3.5 $\pm$ 2.1	5.3 $\pm$ 2.6	0.76 [0.18–1.33]
Jogging differences				
Early ACLR				
vGRF	20%–23%	19.3 $\pm$ 1.3	20.2 $\pm$ 1.0	0.81 [0.423–1.38]
External knee flexion moment	9%–22%	0.60 $\pm$ 0.34	0.91 $\pm$ 0.23	1.18 [0.58–1.78]
Knee flexion motion	9%–25%	29.2 $\pm$ 4.3	33.4 $\pm$ 4.8	0.90 [0.32–1.48]
External knee adduction moment	9%–11%,	0.24 $\pm$ 0.26	0.46 $\pm$ 0.22	0.97 [0.38–1.55]
Mid ACLR (No differences)				
Late ACLR				
External knee flexion moment	8%–18%	0.79 $\pm$ 0.17	0.97 $\pm$ 0.21	0.90 [0.30–1.49]
Knee abduction motion	32%–39%	0.4 $\pm$ 2.2	2.3 $\pm$ 2.6	0.77 [0.19–1.36]
External hip adduction moment	21%–23%	1.33 $\pm$ 0.44	0.94 $\pm$ 0.35	1.04 [0.44–1.64]

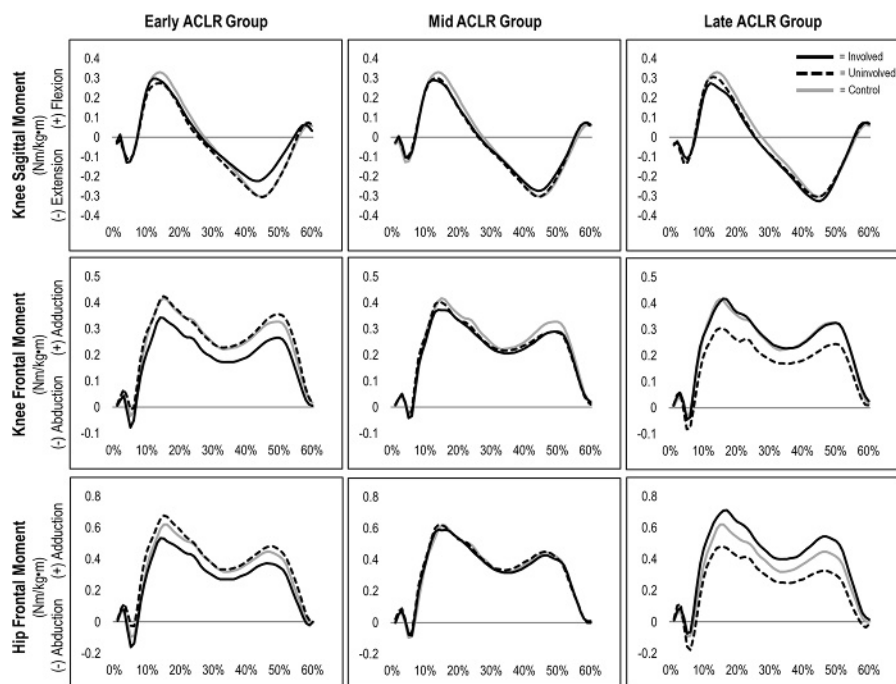


FIGURE 1—Walking gait mean ensemble curves of knee sagittal, knee frontal, and hip frontal moments in the early, mid, and late ACLR groups: involved vs uninjured vs control limbs.

during terminal-swing, and less hip abduction motion during mid-swing compared with the healthy control knees. While jogging, the late ACLR group demonstrated lower external knee flexion moments during loading-response through mid stance, less knee abduction motion during terminal-stance, and greater external hip adduction moments during mid stance on the involved limb compared with the healthy control group.

Comparisons between ACLR groups. The only significant differences observed between the three ACLR groups were between the early ACLR and late ACLR group. While walking, the late ACLR group demonstrated greater external knee extension moments during terminal stance (40%–49%; L, $0.30 \pm 0.10 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$; E, $0.20 \pm 0.10 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$, $d = 0.95 [0.26, 1.64]$), greater external hip adduction

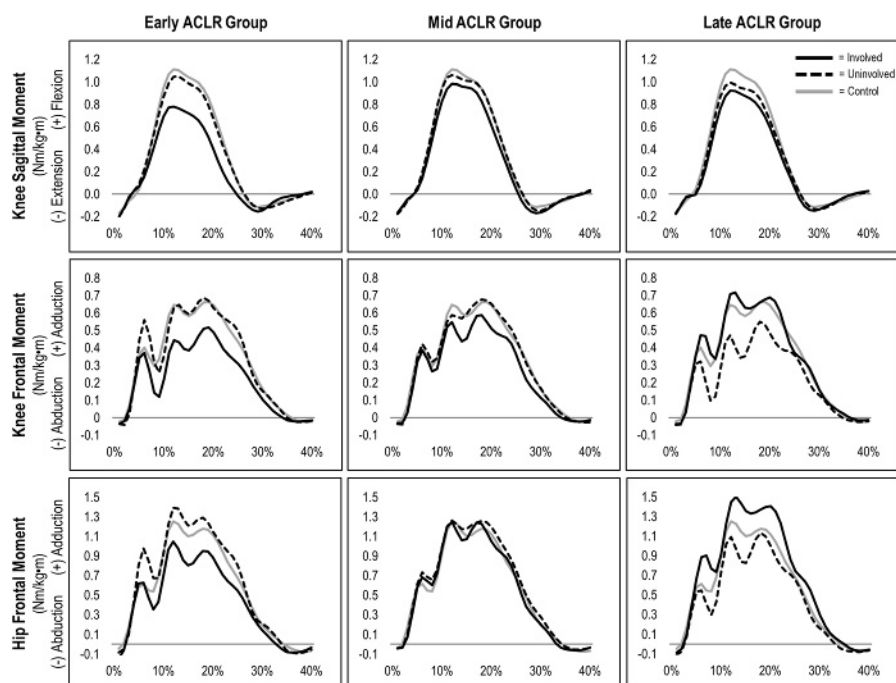


FIGURE 2—jogging gait mean ensemble curves of knee sagittal, knee frontal, and hip frontal moments in the early, mid, and late ACLR groups: involved vs uninjured vs control limbs.

moments during mid and terminal-stance (16%–49%; L: $0.51 \pm 0.15 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$, E: $0.36 \pm 0.20 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$, $d = 0.88$ [0.20–1.57]), and less hip abduction motion during pre, initial, and mid swing (58%–90%, L: $5.2^\circ \pm 2.3^\circ$; E: $7.5^\circ \pm 2.6^\circ$, $d = 0.92$ [0.23–1.61]) compared to the early ACLR group. During the stance phase of jogging, the late ACLR group demonstrated greater external hip adduction moments than the early ACLR group (14%–23%, L: $1.33 \pm 0.42 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$, E: $0.83 \pm 0.52 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}\cdot\text{m}^{-1}$; $d = 1.06$ [0.35, 1.76]).

DISCUSSION

Only the early and late ACLR groups demonstrated gait abnormalities, with both groups displaying gait asymmetries and deviations from healthy controls for multiple variables. Neither the mid ACLR nor healthy control groups demonstrated interlimb differences for any gait variables. Additionally, there were no differences in any gait variables between the mid ACLR and healthy control groups. The early and late ACLR groups were also the only ACLR groups that demonstrated between group differences. All interlimb differences in gait were supported by large magnitude effect-sizes (0.85–1.13) and all between groups differences were supported by moderate to large effect sizes (0.74–1.18). All effect sizes had 95% CI that did not cross zero, suggesting that all observed gait adaptations may be clinically meaningful.

The early ACLR group demonstrated lower sagittal and frontal plane external moments on their involved knee compared to the contralateral or healthy control limbs while walking. These findings are consistent with previous studies that have observed lower peak external knee extension moments in ACLR limbs averaging 9 to 26 months (22,35) postsurgery when compared with contralateral (22) and control (35) knees, and reduced peak external knee adduction moments in ACLR limbs averaging 10 to 26 months postsurgery compared to contralateral knees (20–22). Lower sagittal and frontal knee joint moments and vGRF on the involved limb of the early ACLR group may suggest a protective adaptation in walking gait early after ACLR to limit joint loading and reduce the demand on muscles, such as the quadriceps and hip abductors (11,36,37), that are normally used to attenuate joint loads.

Similar to the walking findings, the early ACLR group demonstrated lower joint loading on the ACLR limb during jogging. This included lower external knee flexion moments and knee flexion motion compared to contralateral and control knees, which are patterns suggestive of “quadriceps avoidance” (16,17). The higher forces associated with jogging may explain the presences of quadriceps avoidance pattern during jogging but not walking in the early ACLR group. Interestingly, despite jogging being a commonly recommended sagittal plane exercise for patients in the early stages after ACLR, there is limited evidence describing jogging biomechanics in individuals early after ACLR (15). This may be an area for further clinical study given the high

forces of jogging and the implications that early adaptations may have on long-term movement strategies.

The most surprising finding of this study was that the mid ACLR group demonstrated symmetrical lower extremity gait patterns similar to the healthy control group for all variables during both walking and jogging. These findings may suggest a “normalization” of gait in this group and are particularly interesting when we consider that both the early ACLR and late ACLR groups demonstrated numerous abnormal joint loading patterns compared to the contralateral and healthy control limbs. Some prior studies have also observed no differences in external knee flexion moments (21), knee extension moments (21), or knee adduction moments (26,38) between ACLR limbs and contralateral or control limbs in participants an average of 3.3 to 4.5 yr postsurgery. While other studies have observed lower external knee flexion moments (26) and knee adduction moments (21,28) compared with contralateral and control knees in participants averaging 3.3 to 3.5 yr postsurgery. Despite the average time postsurgery for all of these prior studies falling within the 2- to 5-yr time frame for our mid ACLR group, most of these studies included participants from broader times postsurgery (6 months to 11 yr), with only one study (21) having a sample with a similar narrow time postsurgery (3.3 ± 0.4 yr) as our mid ACLR group. That study observed no differences in external knee flexion and extension moments, but the external adduction moment was smaller on the ACLR knee compared with the contralateral limb (21).

The late ACLR group demonstrated lower sagittal plane knee joint loading, but greater frontal plane knee and hip joint loading on the ACLR limb during walking and jogging. Lower external knee flexion moments were observed during both walking and jogging when compared to the healthy control limb. One previous study of ACLR participants 5+ yr postsurgery (5.2 ± 3.2 yr) also observed lower external knee flexion moments during walking and a trend toward lower external knee flexion moments during jogging (27), however, another average of 6 yr postsurgery (range, 2–18 yr) reported no differences in external knee flexion or extension moments compared to contralateral or control knees (39). The presence of the “quadriceps avoidance” pattern in the late ACLR group may be related to an inability of the quadriceps muscles to eccentrically absorb joint loads. Theoretically, an inability to adequately attenuate forces on the knee could lead to progressive degradation of articular surfaces (40).

In the frontal plane, the late ACLR group demonstrated greater external knee and hip adduction moments during walking compared with the contralateral limb, and greater external hip adduction moments during jogging compared to the contralateral and healthy control limb. There were no differences in knee adduction moment between the ACLR limb and the control limb, which may suggest that interlimb difference in knee adduction moment were driven by a lower-knee adduction moment in the contralateral limb. During jogging, the knee adduction moment was greater in the ACLR limbs compared with contralateral for multiple nonconsecutive

regions of the gait; however, these regions were each only 2% of the gait cycle and therefore did not meet our criteria for a statistically significant difference. Previous reports of the external knee adduction moments in ACLR samples averaging greater than 5 yr postsurgery have conflicted. One study of participants 5.3 ± 4.4 yr postsurgery observed greater peak external knee adduction moments during walking compared to controls (19), while another study with participant an average of 6 yr postsurgery (range, 2–18 yr) observed no significant difference in external knee adduction moments compared to controls (39). Despite increased clinical and scientific focus on posttraumatic adaptations in gait and joint loading as potential risk factors for the development of OA, there are limited investigations examining gait patterns in samples of ACLR participants that are all 5+ yr postsurgery. The presence of a greater knee adduction moment in only the late ACLR group is an interesting finding in context of the theoretical relationship between the knee adduction moment and the development of knee OA (23,24). Additionally, while it is not a commonly reported variable in the ACLR gait literature, the external hip adduction moment was greater in the late ACLR group for multiple comparisons and may warrant future investigation as a variable of interest.

The findings of the current study may support an alternative to the conventional theory that abnormal gait patterns are a consistently present after ACLR. Aligning with conventional theory, the early ACLR group demonstrated altered knee and hip joint loading compared to contralateral and healthy control limbs. However, the findings that the mid ACLR group showed no altered gait patterns while the late ACLR group demonstrated potentially harmful knee and hip joint loading patterns deviates from conventional theory. Both the early and late ACLR groups demonstrated lower external knee flexion moments and altered knee and hip adduction moments, but not the mid ACL group. While this is a cross-sectional analysis, the theoretical presence of a temporal curve in joint loading from recovery to deterioration could impact how we manage and study ACLR patients over the course of time postsurgery. Patients in this “mid” time frame postsurgery may present clinically as recovered and “normal,” but may still be at risk for experiencing new changes in joint loading as they progress along a greater time postsurgery. Clinical care and research efforts could benefit from including serial follow-ups of ACLR patients to track and treat progressive changes in muscle function and movement patterns that develop later over the course of ACLR chronicity. Although the findings of this study provide supporting evidence for the theory that gait patterns may improve then decline over time postsurgery, future longitudinal research is necessary to substantiate this theory and better understand the role of abnormal gait patterns and the etiology and prevention of PTOA.

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This study is not without limitations. As an initial investigation in this area, potential differences due to time postsurgery were studied using a cross-sectional design limiting our ability to make cause-effect conclusions. Time postsurgery cutoffs for stratification were based on clinically relevant time frames: <2 yr, a time frame most patients are returning to sport and at are highest risk of suffering a secondary ACL injury (41), 2 to 5 yr, a time frame patients are outside the high-risk window and should be back to activity, and 5 to 15 yr, a long-term time frame postsurgery but theoretically before major joint degeneration. Statistical comparisons across the gait cycle allowed for more comprehensive comparisons between limbs and groups; however, a limitation of this analysis technique is that it does not account for potential statistical error associated with multiple comparisons. Radiographic examinations were not performed to definitively know whether participants were already developing signs of knee OA. This makes it particularly difficult to theorize whether gait adaptations observed in the late ACLR group are a precursor to or a product of developing knee OA. We attempted to maintain homogeneity of demographics between ACLR groups outside of the time postsurgery variable; however, although the average age of the late ACLR group was only 26.7 yr, they were significantly older than the other groups. Additionally, surgical procedures, graft types, postsurgical rehabilitation, and joint laxity were not controlled for between groups. Based on the evidence suggesting that ACLR tunnel orientation (42) and meniscal involvement (43) can alter knee joint kinetics and kinematics, not controlling for these factors may have influenced our findings.

CONCLUSIONS

Walking and jogging gait biomechanics presented differently in patients at different stages in time after ACLR surgery. The early ACLR group demonstrated lower sagittal and frontal plane joint loading on the ACLR limb compared with contralateral and control limbs. The mid ACLR group did not demonstrate any gait differences compared with the contralateral or control limb. The late ACLR group demonstrated lower sagittal plane joint loading compared with control limbs and greater frontal plane joint loading compared with contralateral and control limbs.

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